卡尔·雅可比（Carl Gustav Jacob Jacobi，1804~1851）**,德国**数学家。

5.2 雅可比迭代法

提要

- Jacobi 迭代法举例
- Jacobi 迭代法的迭代公式

例 5.2.1　用雅可比迭代法求解线性方程组。

$$\begin{cases} 10x_1 - x_2 - 2x_3 = 72 \\ -x_1 + 10x_2 - 2x_3 = 83 \\ -x_1 - x_2 + 5x_3 = 42 \end{cases}$$

$$\begin{cases} x_1^{(k+1)} = \dfrac{1}{10}\left(x_2^{(k)} + 2x_3^{(k)} + 72\right) \\ x_2^{(k+1)} = \dfrac{1}{10}\left(x_1^{(k)} + 2x_3^{(k)} + 83\right) \\ x_3^{(k+1)} = \dfrac{1}{5}\left(x_1^{(k)} + x_2^{(k)} + 42\right) \end{cases}$$

$$x^{(k+1)} = \begin{pmatrix} 0 & 0.1 & 0.2 \\ 0.1 & 0 & 0.2 \\ 0.2 & 0.2 & 0 \end{pmatrix} x^{(k)} + \begin{pmatrix} 7.2 \\ 8.3 \\ 8.4 \end{pmatrix}$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0.0000	0.0000	0.0000	5	10.9510	11.9510	12.9414
1	7.2000	8.3000	8.4000	6	10.9834	11.9834	12.9504
2	9.7100	10.7000	11.5000	7	10.9944	11.9981	12.9934
3	10.5700	11.5700	12.4820	8	10.9981	11.9941	12.9978
4	10.8525	11.8534	12.8282	9	10.9994	11.9994	12.9992

雅可比迭代法公式推导
（线性方程组的表示形式）

$$\sum_{j=1}^{n} a_{ij} x_j = b_i \qquad (i = 1, 2, \cdots, n)$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

雅可比迭代法公式-分量形式

$$\sum_{j=1}^{n} a_{ij} x_j = b_i \qquad (i = 1, 2, \cdots, n)$$

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1, j \neq i}^{n} a_{ij} x_j \right) \qquad (i = 1, 2, \cdots, n)$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(- \sum_{j=1, j \neq i}^{n} a_{ij} x_j^{(k)} + b_i \right)$$

雅可比迭代法公式-矩阵形式

$$x^{(k+1)} = B_{\mathrm{J}} x^{(k)} + f_{\mathrm{J}}$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

若将系数矩阵 A 分解为 $A = D - L - U$，

$$D = \begin{pmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{pmatrix} \quad -L = \begin{pmatrix} 0 & & & & \\ a_{21} & 0 & & & \\ a_{31} & a_{32} & 0 & & \\ \vdots & \vdots & \ddots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn-1} & 0 \end{pmatrix} \quad -U = \begin{pmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ & 0 & a_{23} & \cdots & a_{2n} \\ & & 0 & \ddots & \vdots \\ & & & \ddots & a_{n-1n} \\ & & & & 0 \end{pmatrix}$$

$$(D - L - U)x = b \qquad Dx = (L + U)x + b$$

$$x = D^{-1}(L+U)x + D^{-1}b = D^{-1}(D - A)x + D^{-1}b$$

$$= \left(I - D^{-1}A\right)x + D^{-1}b = B_J x + f_J$$

雅可比迭代法公式-矩阵形式

$$Ax = b$$

$$x^{(k+1)} = B_J x^{(k)} + f_J$$

$$B_J = I - D^{-1}A \qquad f_J = D^{-1}b$$

$$= D^{-1}(L+U)$$

例 5.2.1 用雅可比迭代法求解线性方程组。

$$\begin{cases} 10x_1 - x_2 - 2x_3 = 72 \\ -x_1 + 10x_2 - 2x_3 = 83 \\ -x_1 - x_2 + 5x_3 = 42 \end{cases}$$

$$\boldsymbol{B}_{\mathrm{J}} = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A}$$

$$= \boldsymbol{D}^{-1}(\boldsymbol{L}+\boldsymbol{U})$$

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{B}_{\mathrm{J}}\boldsymbol{x}^{(k)} + \boldsymbol{f}_{\mathrm{J}}$$

$$\boldsymbol{f}_{\mathrm{J}} = \boldsymbol{D}^{-1}\boldsymbol{b}$$

$$\boldsymbol{x}^{(k+1)} = \begin{pmatrix} 0 & 0.1 & 0.2 \\ 0.1 & 0 & 0.2 \\ 0.2 & 0.2 & 0 \end{pmatrix} \boldsymbol{x}^{(k)} + \begin{pmatrix} 7.2 \\ 8.3 \\ 8.4 \end{pmatrix}$$

$$\begin{cases} x_1^{(k+1)} = \dfrac{1}{10}\left(x_2^{(k)} + 2x_3^{(k)} + 72\right) \\ x_2^{(k+1)} = \dfrac{1}{10}\left(x_1^{(k)} + 2x_3^{(k)} + 83\right) \\ x_3^{(k+1)} = \dfrac{1}{5}\left(x_1^{(k)} + x_2^{(k)} + 42\right) \end{cases}$$

- End.